Time: 12-3 PM

Name:

Student ID:

Q1: Choose the correct answer for each of the following questions. (Each question has one correct answer)

1. **What is the primary goal of a Conversational IR system?**
   A. To entertain the user through jokes
   B. To retrieve relevant information in a multi-turn dialogue
   C. To summarize all online content
   D. To translate user input into multiple languages
2. **Why is query rewriting important in a conversational IR system like TREC CAsT?**
   A. It reduces server load
   B. It generates synthetic data for training
   C. It reformulates ambiguous user queries using dialogue context
   D. It translates queries into different languages
3. **In the TREC CAsT dataset, how is the user's current query clarified during the evaluation process?**
   A. By translating it to French
   B. By using retrieved answers from a knowledge base
   C. By providing a manually rewritten query with context
   D. By removing stop words from the query
4. **Which of the following datasets is designed for multi-turn conversational QA?**
   A. TREC-COVID
   B. CoQA
   C. WikiQA
   D. CNN-DailyMail
5. **What does "contextual dependency" refer to in CIR?**
   A. Linking queries to external APIs
   B. Using the history of conversation to interpret new queries
   C. Grouping multiple queries into one
   D. Detecting the user's geographical location

6. **Which model type is best suited for contextual query rewriting in CIR?**
   A. CNN
   B. RNN
   C. Transformer-based model (e.g., T5 or BERT)
   D. Logistic Regression
7. **What does TREC stand for in IR research?**
   A. Text Retrieval Evaluation Conference
   B. Technical Review of Educational Content
   C. Training for Robotic Entity Conversations
   D. Testing Recursive Entity Classification
8. **Which of the following best describes the challenge of ambiguous queries in CIR?**
   A. Queries that are too long
   B. Queries that refer to previous conversation but lack context
   C. Queries about recent news
   D. Queries containing code
9. **Which of the following characteristics is essential for a CIR system to maintain user engagement over long-term conversations, as demonstrated in XiaoIce?**
   A. Frequent advertisements
   B. Avoiding retrieval-based answers
   C. Building emotional connection and personality consistency
   D. Asking the user to rephrase all queries manually
10. **Which of the following is NOT a factor used in topic selection for dialogue agents like XiaoIce?**
    A. Freshness
    B. Contextual relevance
    C. User's keyboard language
    D. Popularity

Q2: Essay questions

1. Clearly describe the process of **contextual query rewriting** in Conversational Information Retrieval systems.
   In your answer, explain how such systems resolve ambiguity caused by **ellipsis, coreference**, and **omission**, and provide relevant examples.

2. Define **conversational dependencies** within **multi-turn retrieval** settings, particularly in the context of **TREC CAsT Year 2**.
   Why are these dependencies important, and how do they affect system performance? Support your explanation with examples.

3. The **CoQA dataset** has become a benchmark in conversational question answering.
   Discuss its key features, and explain the **main challenges** that arise when using CoQA in CIR research and system development.

4. Design a complete **Conversational IR pipeline** using the **Cambridge four-stage model**.
   Your pipeline should include the following components: **query rewriting, passage retrieval, reranking, and answer generation**.
   Briefly explain the role of each component in improving conversational search

5. In Conversational IR, **clarifying user intent** is essential for accurate and helpful responses.
   Discuss this importance and compare **two major approaches** to intent clarification: **Abg-CoQA** and **ClarQ**.

Q3: Practical questions

Using a sample topic from the **TREC CAsT 2020 dataset** (e.g., Topic 24 or Topic 34), write a Python function that identifies **ambiguous user queries** within a multi-turn conversation. function must:

- Analyze each user query with its **preceding context (2-3 previous turns)**.
- Detect ambiguity types such as **coreference, ellipsis, or incomplete references**.

use libraries like: transformers (from HuggingFace) , spaCy or nltk (for basic NLP)

output should include:

1. The original conversation turn
2. A boolean flag: ambiguous / not ambiguous
3. Explanation of the ambiguity
4. Rewritten query\

(You may optionally use bert-base-uncased or another pretrained transformer model.)

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